# Resourcing

## Overview

Analyzing questions related to future prospects for OSPOs.

Research questions:
What does the next two years hold for OSPOs?
Where are OSPO staff working to grow?

Survey questions:
Over the next two years, what do you expect to happen to your OSPO or OSPO-like function?
What is the time horizon of your current funding or allocation?
Which areas do you expect to become more important for your OSPO over the next two years?
Which area would you most like to grow, and what is stopping you?
What skill or expertise do you most wish you or your OSPO had more of?

## Import packages, utilities, and config variables

```
library(readxl)
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag
```

```
The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
library(tibble)
```

```
Warning: package 'tibble' was built under R version 4.4.3
```

```
library(readr)
```

```
Warning: package 'readr' was built under R version 4.4.3
```

```
library(tidyr)
```

```
Warning: package 'tidyr' was built under R version 4.4.3
```

```
source("utils.R")

# Read config.yml
cfg <- config::get()
```

## Load data

```
mc_simple <- read_tsv(file.path(cfg$data_path, "mc_simple.tsv"), show_col_types = FALSE, ski
org_responses <- read_excel(file.path(cfg$curated_path, "orgs.xlsx"), sheet="responses")
future <- read_tsv(file.path(cfg$data_path, "future_areas.tsv"), show_col_types = FALSE, ski
ospo_loc <- read_tsv(file.path(cfg$data_path, "ospo_location.tsv"), show_col_types = FALSE, 
codebook <- read_excel(file.path(cfg$data_path, "codebook.xlsx"))
mc_complex <- read_tsv(file.path(cfg$data_path, "mc_complex.tsv"), show_col_types = FALSE, sk
qual <- read_tsv(file.path(cfg$data_path, "qual.tsv"), show_col_types = FALSE, skip_empty_rou
```

## What does the next two years hold for OSPOs?

```r
table(mc_simple$expectations)
```

```
                  At risk of ending Grow (more staff, funding, or scope)
                                  2                                    14
                             Shrink                  Stay about the same
                                  4                                    12
```

```r
table(mc_simple$funding_duration)
```

```
                                    Annual
                                         3
   Multi-year (2 or more years remaining)
                                         3
                       No defined funding
                                         7
Ongoing or recurring institutional budget
                                         8
      Tied to a grant or project end date
                                        11
```

Just for my own reassurance, I want to run some chi-square tests on some variables that I suspect might be correlated.

```r
# Expectations vs Funding
chisq.test(mc_simple$expectations, mc_simple$funding_duration)
```

```
Warning in chisq.test(mc_simple$expectations, mc_simple$funding_duration):
Chi-squared approximation may be incorrect
```

```

    Pearson's Chi-squared test

data:  mc_simple$expectations and mc_simple$funding_duration
X-squared = 12.955, df = 12, p-value = 0.3723
```

```r
# Expectations vs Formalization
chisq.test(mc_simple$expectations, mc_complex$function_mc)
```

```
Warning in chisq.test(mc_simple$expectations, mc_complex$function_mc):
Chi-squared approximation may be incorrect
```

```

    Pearson's Chi-squared test

data:  mc_simple$expectations and mc_complex$function_mc
X-squared = 5.6904, df = 9, p-value = 0.7705
```

```r
# Funding vs Formalization
chisq.test(mc_simple$funding_duration, mc_complex$function_mc)
```

```
Warning in chisq.test(mc_simple$funding_duration, mc_complex$function_mc):
Chi-squared approximation may be incorrect
```

```

    Pearson's Chi-squared test

data:  mc_simple$funding_duration and mc_complex$function_mc
X-squared = 19.929, df = 12, p-value = 0.06844
```

Well, the last one is bordering on significant, but it's the least interesting of the three relationships.

```r
tmp <- cbind(mc_simple$expectations, ospo_loc)
names(tmp)[1] <- "expectations"
tmp |> filter(tto == "Yes")
```

```
                        expectations library admin tto it hpc rse_group
1                  Stay about the same     Yes    No Yes No Yes        No
2                  Stay about the same      No    No Yes No  No        No
3 Grow (more staff, funding, or scope)      No    No Yes No  No        No
4                  Stay about the same      No    No Yes No  No        No
5 Grow (more staff, funding, or scope)      No    No Yes No  No        No
6 Grow (more staff, funding, or scope)      No    No Yes No  No        No
7 Grow (more staff, funding, or scope)      No    No Yes No  No        No
```

```
8                  Stay about the same      No    No Yes No  No        No
9 Grow (more staff, funding, or scope)      No    No Yes No  No        No
  academic_dept research_center ai_institute joint
1            No              Yes           No   Yes
2            No               No           No    No
3            No               No           No    No
4            No               No           No    No
5            No               No           No   Yes
6            No               No           No    No
7            No               No           No    No
8            No               No           No    No
9            No               No           No    No
```

```
tmp |> filter(library == "Yes")
```

```
                         expectations library admin tto it hpc rse_group
1                  Stay about the same     Yes    No Yes No Yes        No
2                               Shrink     Yes    No  No No  No        No
3  Grow (more staff, funding, or scope)     Yes    No  No No  No        No
4  Grow (more staff, funding, or scope)     Yes    No  No No  No        No
5                  Stay about the same     Yes    No  No No  No        No
6                  Stay about the same     Yes    No  No No  No        No
7                    At risk of ending     Yes    No  No No  No        No
8                               Shrink     Yes    No  No No  No        No
9                                 <NA>     Yes    No  No No  No        No
10 Grow (more staff, funding, or scope)     Yes    No  No No  No        No
11                 Stay about the same     Yes    No  No No  No        No
   academic_dept research_center ai_institute joint
1            No              Yes           No   Yes
2            No               No           No    No
3            No               No           No    No
4            No               No           No    No
5            No               No           No    No
6            No               No           No    No
7            No               No           No    No
8            No               No           No    No
9            No               No           No    No
10           No               No           No    No
11           No               No           No    No
```

Hmm. I'm forming a hypothesis that OSPOs associated with a TTO are in better shape than those associated with libraries. Let's create an "optimism" score. "Optimists" this their

department will either grow or stay the same.

```
tmp2 <- tmp |>
  mutate(
    optimistic = expectations %in% c("Grow (more staff, funding, or scope)", "Stay about the
    group = ifelse(tto == "Yes", "TTO", "Non-TTO")
  )|>
    select(c(optimistic, group))

head(tmp2)
```

```
  optimistic   group
1       TRUE     TTO
2       TRUE Non-TTO
3      FALSE Non-TTO
4       TRUE     TTO
5       TRUE     TTO
6      FALSE Non-TTO
```

```
tmp2 |>
  group_by(group) |>
  summarise(
    n = n(),
    optimistic_count = sum(optimistic, na.rm = TRUE),
    pct_optimistic = round(100 * mean(optimistic, na.rm = TRUE), 1)
  )
```

```
# A tibble: 2 x 4
  group       n optimistic_count pct_optimistic
  <chr>   <int>            <int>          <dbl>
1 Non-TTO    24               17           70.8
2 TTO         9                9          100
```

Meh, not sure if that's a significant trend. Let's test it.

```
chisq.test(tmp2$group, tmp2$optimistic)
```

```
Warning in chisq.test(tmp2$group, tmp2$optimistic): Chi-squared approximation
may be incorrect
```

```
    Pearson's Chi-squared test with Yates' continuity correction

data:  tmp2$group and tmp2$optimistic
X-squared = 1.8151, df = 1, p-value = 0.1779
```

Nah.

### What areas will be important for OSPOs in the coming years?

I see one person gave all NA responses to this question. I will remove them. Nobody else gave any NAs.

```
# N.B. this removes rows with NA in ANY column
dim(future)
```

```
[1] 33  8
```

```
future <- na.omit(future)
dim(future)
```

```
[1] 32  8
```

```
count_yes_responses(future)
```

```
# A tibble: 8 x 3
  count tip                                                          nickname
  <dbl> <chr>                                                        <chr>
1    27 AI and open source (policy, guidance, tooling)                future_ai
2    22 Research software sustainability and long-term maintenance    future_s~
3    22 Demonstrating impact and value to leadership                  future_i~
4    15 Cross-institutional collaboration (shared services, consortia) future_c~
5    14 Security and supply chain requirements                        future_s~
6    14 Commercialization and industry partnership                    future_c~
7    13 Training and skills development                               future_t~
8    11 Open science and funder compliance                            future_o~
```

## Where are OSPO staff working to grow?

Sending the last two questions to file for qualitative coding.

```
write_df_to_file(qual$professional_growth, "professional_growth.txt", path_key = "coding_pat
write_df_to_file(qual$desired_skills, "desired_skills.txt", path_key = "coding_path")
```